

A graded Pieri decomposition for the rectangular Δ -Springer ring $R_{2k,(k,k),2}$

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Abstract

For each $k \geq 1$, the graded Frobenius character of the Chou–Hanada Δ -Springer ring $R_{2k,(k,k),2}$ (arXiv 2509.24252) is given by the one-term-per-shape formula

$$\mathrm{Frob}_q(R_{2k,(k,k),2}) = \sum_{j=0}^k q^j s_{(2k-j,j)}.$$

Each two-row Schur function $s_{(2k-j,j)}$ appears with multiplicity one and in its own graded degree j . This is a q -graded lift of the classical multiplicity-free branching $\mathrm{Ind}_{S_k \wr S_2}^{S_{2k}} \mathbf{1} = \bigoplus_{j=0}^k S^{(2k-j,j)}$. The proof is a direct enumeration of the Griffin–Chou–Hanada higher-Specht index set at rectangular parameters: the constraints $\mathrm{des}(S) \leq 1$ and $\mathrm{ctype}(S) \supseteq (k,k)$ together admit exactly one standard Young tableau per two-row shape, and Blasiak’s catabolism algorithm identifies its cocharge with the natural row-jump statistic.

1 Setup

We work with the Chou–Hanada Δ -Springer ring $R_{n,\lambda,s}$ of arXiv 2509.24252 and its graded Frobenius character. All conventions match theirs (and Griffin, arXiv 2004.00788): tableaux are drawn in the French convention, so row indices increase from bottom to top, and columns strictly increase in that direction.

Let $n \geq 1$, $\lambda \vdash k$ with $\ell(\lambda) \leq s$. Denote by $\mathrm{SYT}(n)$ the set of standard Young tableaux with entries $\{1, \dots, n\}$. For $S \in \mathrm{SYT}(n)$ and $t \leq n$, write $S|_t$ for the skew SYT obtained by deleting the cells of S with entries $> t$.

Definition 1 (Cocharge; CH Def. 2.14). Given $S \in \mathrm{SYT}(n)$, the cocharge tableau $\mathrm{cc}(S)$ assigns the label 0 to the cell containing 1; and inductively, for $i = 1, \dots, n-1$, the cell containing $i+1$ receives the same label as i if $i+1$ lies in the same or a strictly lower row than i , and one more otherwise. The cocharge of S is $\mathrm{cocharge}(S) := \sum \mathrm{cc}(S)$.

Definition 2 (Descents). For $S \in \mathrm{SYT}(n)$, define $\mathrm{des}(S)$ as the number of indices $i \in \{1, \dots, n-1\}$ such that $i+1$ lies in a strictly higher row than i . Equivalently (Chou–Hanada), $\mathrm{des}(S) = \max \mathrm{cc}(S)$.

Definition 3 (Catabolisability type; Blasiak, CH Alg. 1). Given $S \in \mathrm{SYT}(n)$, form the reading word $\mathrm{rw}(S)$ by concatenating rows top-to-bottom, each left-to-right, and the cocharge word $z = z_1 z_2 \cdots z_n$ by replacing each entry v in $\mathrm{rw}(S)$ by its cocharge label $\mathrm{cc}(S)_v$. Starting with the empty partition $\nu = \emptyset$, iterate the following step on the pair (x, ν) initially equal to $(z, \emptyset)$: let a be the last letter of x . If adding one box to row $a+1$ of ν leaves ν a partition, delete a from x and perform

that addition; otherwise, remove a from the end of x and prepend $a + 1$ to x . Terminate when x is empty. The resulting ν is $\text{ctype}(S)$.

The Chou–Hanada higher-Specht index set is

$$\Sigma_{n,\lambda,s} = \{(S, \mu) : S \in \text{SYT}(n), \text{ctype}(S|_k) \supseteq \lambda, \text{des}(S) < s, \mu \subseteq (n-k) \times (s - \text{des}(S) - 1)\},$$

where $k = |\lambda|$ and $\mu \subseteq (a) \times (b)$ means $\ell(\mu) \leq a$ and $\mu_1 \leq b$. The relation $\supseteq$ is dominance order.

Theorem 4 (Chou–Hanada, Theorem B & Cor. 2.33, arXiv 2509.24252).

$$\text{Frob}_q(R_{n,\lambda,s}) = \sum_{(S,\mu) \in \Sigma_{n,\lambda,s}} q^{\text{cocharge}(S)+|\mu|} s_{\text{sh}(S)}.$$

We take Theorem 4 as our starting point. Our formula is a consequence of specialising it at rectangular parameters.

2 Main theorem

Theorem 5. *For every integer $k \geq 1$,*

$$\text{Frob}_q(R_{2k,(k,k),2}) = \sum_{j=0}^k q^j s_{(2k-j,j)}.$$

Remark 6. Ungraded, $\sum_{j=0}^k f^{(2k-j,j)} = \binom{2k}{k}$ by the standard enumeration of two-row SYT; this matches $\dim R_{2k,(k,k),2} = (2k)!/(k!)^2$ from Chou–Hanada Cor. 2.34.

Reduction of the index set

Fix $(n, \lambda, s) = (2k, (k, k), 2)$. Then $|\lambda| = 2k = n$, so $S|_{|\lambda|} = S$ for every $S \in \text{SYT}(n)$, and the rectangle in the second slot has $n - |\lambda| = 0$ rows, forcing $\mu = \emptyset$. Substituting into Theorem 4,

$$\text{Frob}_q(R_{2k,(k,k),2}) = \sum_{\substack{S \in \text{SYT}(2k) \\ \text{des}(S) \leq 1 \\ \text{ctype}(S) \supseteq (k,k)}} q^{\text{cocharge}(S)} s_{\text{sh}(S)}. \quad (1)$$

The proof of Theorem 5 amounts to describing this restricted tableau set explicitly.

Row bound from the descent constraint

Lemma 7. *For any $S \in \text{SYT}(n)$ of shape ν , $\text{des}(S) \geq \ell(\nu) - 1$.*

Proof. Let $\ell = \ell(\nu)$ and let v_r denote the entry in column 1, row r of S , for $r = 1, \dots, \ell$. Since columns of S strictly increase from bottom to top, $v_1 < v_2 < \dots < v_\ell$. Fix $r \in \{2, \dots, \ell\}$. The entry v_r is the leftmost cell of row r , hence the smallest entry in that row, so $v_r - 1$ does not lie in row r . If $v_r - 1$ lay in a row $r' > r$, then the leftmost cell of row r' would contain some value $v_{r'} \leq v_r - 1 < v_r$, contradicting $v_r < v_{r'}$ (column-1 strict increase). Therefore $v_r - 1$ lies in a row strictly below r , and v_r (which is in row r) sits in a strictly higher row than $v_r - 1$. This contributes a descent at position $v_r - 1$. As r varies over $\{2, \dots, \ell\}$, the resulting positions $v_r - 1$ are pairwise distinct, giving $\ell - 1$ distinct descents. $\square$

Corollary 8. *If $S \in \text{SYT}(2k)$ satisfies $\text{des}(S) \leq 1$, then $\text{sh}(S) = (2k - j, j)$ for some $0 \leq j \leq k$.*

Proof. By Lemma 7, $\ell(\text{sh}(S)) \leq 2$. As $|\text{sh}(S)| = 2k$, the shape is $(2k - j, j)$ with $0 \leq j \leq k$. $\square$

Uniqueness of the qualifying SYT

For $0 \leq j \leq k$, define the two-row SYT S_j of shape $(2k - j, j)$ by placing $1, 2, \dots, 2k - j$ in the bottom row (row 1) left-to-right and $2k - j + 1, 2k - j + 2, \dots, 2k$ in the top row (row 2) left-to-right:

$$S_j = \boxed{2k-j+1} \boxed{2k-j+2} \cdots \boxed{2k} / \boxed{1} \boxed{2} \cdots \cdots \cdots \boxed{2k-j}$$

(top row above bottom row, French convention). This is a valid SYT: each column $c \in \{1, \dots, j\}$ has c below $2k - j + c$, and $c < 2k - j + c$.

Lemma 9. *For each $0 \leq j \leq k$, the unique $S \in \text{SYT}(2k)$ of shape $(2k - j, j)$ with $\text{des}(S) \leq 1$ is $S = S_j$. Moreover $\text{des}(S_0) = 0$ and $\text{des}(S_j) = 1$ for $j \geq 1$.*

Proof. For a two-row SYT, a descent at position i means i lies in row 1 and $i + 1$ in row 2.

If $j = 0$: the only SYT of shape $(2k)$ is $1 2 \cdots 2k$, which has no descent, and coincides with S_0 .

Fix $j \geq 1$. By Lemma 7, $\text{des}(S) \geq 1$, so $\text{des}(S) \leq 1$ forces $\text{des}(S) = 1$; call the unique descent position d . Then d lies in row 1 and $d + 1$ in row 2.

We claim $\{1, 2, \dots, d\}$ all lie in row 1: otherwise let $i < d$ be smallest with i in row 2; then $i - 1$ is in row 1 (or $i = 1$, but 1 lies in row 1 of any SYT), so $i - 1$ is a descent position distinct from d , contradicting uniqueness.

Similarly $\{d + 1, d + 2, \dots, 2k\}$ all lie in row 2: suppose some $i > d + 1$ lies in row 1. Since row 2 has size $j \geq 1$, entries $d + 1, \dots, 2k$ must together fill the remaining $2k - d$ cells of the tableau, and some subsequent entry $i' > i$ must jump back up to row 2 (since not all of $\{d + 1, \dots, 2k\}$ can lie in row 1). Take i' to be the smallest such; then $i' - 1$ lies in row 1 and i' in row 2, producing a second descent at $i' - 1 \neq d$. Contradiction.

Row 2 therefore has exactly $2k - d$ entries. As row 2 also has size j by shape, $d = 2k - j$, and S equals S_j . The single descent lies at position $2k - j$. $\square$

Cocharge and catabolism of S_j

Lemma 10. $\text{cocharge}(S_j) = j$ for all $0 \leq j \leq k$.

Proof. Directly from Definition 1. The label of 1 is 0. For $i = 2, \dots, 2k - j$: both $i - 1$ and i lie in row 1, so i is in the same row as $i - 1$ and its label matches, giving $\text{cc}(S_j)_i = 0$. The entry $2k - j + 1$ lies in row 2, strictly higher than $2k - j$ in row 1, so $\text{cc}(S_j)_{2k-j+1} = 1$. For $i = 2k - j + 2, \dots, 2k$: both $i - 1$ and i lie in row 2, so the label remains 1. Summing, $\text{cocharge}(S_j) = j$. $\square$

Lemma 11. $\text{ctype}(S_j) = (2k - j, j)$ for all $0 \leq j \leq k$; equivalently $\text{ctype}(S_j) \supseteq (k, k)$.

Proof. The reading word of S_j (top row first, each row left-to-right; French convention) is

$$\text{rw}(S_j) = (2k - j + 1, 2k - j + 2, \dots, 2k, 1, 2, \dots, 2k - j).$$

By Lemma 10 the cocharge labels are $\text{cc}(S_j)_i = 0$ for $i \leq 2k - j$ and $\text{cc}(S_j)_i = 1$ for $i > 2k - j$. Substituting these into the reading word, the cocharge word is

$$z = (\underbrace{1, 1, \dots, 1}_j, \underbrace{0, 0, \dots, 0}_{2k-j}).$$

Now apply Definition 3 to $(z, \emptyset)$, reading from the right end.

Phase 1 ($2k-j$ zeros). While the last letter is 0: the algorithm would add a box to row $0+1=1$ of ν , which is always legal since row 1 of a partition has no upper bound. Delete the last letter, increment ν_1 by one. Repeat $2k-j$ times to obtain $\nu = (2k-j)$ and $x = (1, 1, \dots, 1)$ (j ones).

Phase 2 (j ones). While the last letter is 1: the algorithm would add a box to row $1+1=2$ of ν . This is legal iff $\nu_2 + 1 \leq \nu_1$, i.e., iff the current $\nu_2 < \nu_1 = 2k-j$. We repeat this addition j times, producing $\nu = (2k-j, m)$ with m growing from 1 to j . The legality condition $m \leq 2k-j$ holds throughout because $j \leq k \leq 2k-j$ (the last inequality is $j \leq k$).

After Phase 2, x is empty and $\nu = (2k-j, j)$. Hence $\text{ctype}(S_j) = (2k-j, j)$.

Dominance: $(2k-j, j) \supseteq (k, k)$ iff $2k-j \geq k$ (first partial sums) and $(2k-j) + j = 2k \geq 2k$ (second partial sums, equality). The first inequality is $j \leq k$, which holds throughout the range. $\square$

Assembling the proof

Proof of Theorem 5. By Corollary 8, any tableau contributing to (1) has shape $(2k-j, j)$ for some $0 \leq j \leq k$. By Lemma 9, the unique such tableau with $\text{des} \leq 1$ is S_j ; by Lemma 11 it satisfies $\text{ctype}(S_j) \supseteq (k, k)$; and by Lemma 10, it contributes $q^j s_{(2k-j, j)}$ to (1). Summing over j ,

$$\text{Frob}_q(R_{2k, (k, k), 2}) = \sum_{j=0}^k q^j s_{(2k-j, j)}. \quad \square$$

3 Verification

The formula was verified by exhaustive Sage-free Python enumeration of the higher-Specht index set (Theorem 4) for $k = 2, 3, 4, 5$ (corresponding to $n = 4, 6, 8, 10$). Enumeration reproduces the target identity term-by-term and matches the ungraded dimensions $\dim R_{2k, (k, k), 2} = \binom{2k}{k} = 6, 20, 70, 252$.

The probe implementation lives at

`~/projects/probes/2026-08-07-chou-hanada-graded-frobenius/probe.py`

and includes calibration against Chou–Hanada Example 2.40, the sanity check $R_n = R_{n, (1^n), n}$ at $n = 3$, and additional rectangular parameters $(r, k) = (2, 3), (3, 2), (3, 3)$; all pass.

4 Discussion

Ungraded content: multiplicity-free branching

At $q = 1$, Theorem 5 recovers the well-known multiplicity-free decomposition

$$\text{Ind}_{S_k \wr S_2}^{S_{2k}}(\mathbf{1}) = \bigoplus_{j=0}^k S^{(2k-j, j)},$$

whose Frobenius image is $\sum_j s_{(2k-j, j)}$, obtainable via Pieri from $h_k \cdot h_k = \sum_{j=0}^k s_{(2k-j, j)}$. Theorem 5 places each irreducible summand $S^{(2k-j, j)}$ in graded degree equal to its displacement j from the trivial summand $S^{(2k)}$.

Aesthetic: the grading is Bruhat step

The two-row shape $(2k - j, j)$ is the unique partition of $2k$ at Bruhat distance j from $(2k)$ within the two-row filter. Its degree in $\text{Frob}_q(R_{2k, (k, k), 2})$ agrees with this distance. Equivalently, in the cocharge model, the qualifying tableau S_j is the “minimal-descent” SYT of shape $(2k - j, j)$: the tableau attaining the descent lower bound of Lemma 7. The grading is measured by how far the leading row-2 entry has shifted from the beginning of the word — a single-block displacement.

Rank two is special

At the next rank $r = 3$, i.e. $(n, \lambda, s) = (3k, (k, k, k), 3)$, the graded Frobenius is no longer one-Specht-per-shape. Direct enumeration at $(r, k) = (3, 2)$ produces polynomial multiplicities of degree > 1 ; for example the coefficient of $s_{(5, 1)}$ in $\text{Frob}_q(R_{6, (2, 2, 2), 3})$ is $q + q^2$. The rank-2 formula is a consequence of the descent bound $\text{des}(S) \leq 1$ pinning shapes to $\ell \leq 2$; at rank $r \geq 3$ shapes with up to r rows contribute, and multiple SYT per shape survive the ctype filter.

Position vs. the composite- d project

This theorem addresses only the character-level (equivalently, ungraded) plus a single grading; it is orthogonal to the composite- d regular-element project which lives on the coinvariant-scale ambient module H_n rather than the parabolic-quotient $R_{n, \lambda, s}$. That module realises composite $d = e/2 \cdot 2$ specialisations via cyclic action of a Coxeter regular element; the present result is a q -analogue of a distinct multiplicity-free branching identity, and neither generalises the other.